

Informations about ESI :

The National School of Computer Science (ESI), formerly known as the National Institute of Computer Science (INI), is an Algerian higher education institution that trains State engineers in computer science. It is located in Oued Smar, about 15 km from the city center of the capital, Algiers, in Algeria.

Founded as the Center for Computer Science Studies and Research (CERI) under the supervision of the former Ministry of Planning and Regional Development, it was renamed and placed under the supervision of the Ministry of Higher Education and Scientific Research in 1984.

CERI was the first specialized computer science training center in Africa. It trained the first State engineers, application engineers, and programmers in Algeria and other African countries (Benin, Cameroon, Gabon, Mali, Mauritania, Senegal, Tunisia, etc.).

Admission:

The National School of Computer Science offers a preparatory class in Computer Science, directly accessible after the baccalaureate, with places reserved exclusively for Algerian students. The school is considered to have the most selective training in Algeria, with approximately 250 to 350 spots available each year, and among the highest admission averages in the country (2025: Mathematics 18.19/20, Science 18.55/20, Technical Mathematics 18.93/20). Students with grades that allow them to enter the school represent 1 in 1000 (0.1%) of the baccalaureate candidates.

What are the negative aspects of the school ESI ?

Some students may struggle with loneliness or social adjustment, especially if they are far from home or in a large, impersonal campus environment, Slow Adaptation and a lot of Pressure , that's why joining a club can be a great experience for many reasons where you are going to meet new People , Enhance your Skills and have much more confidence and practical experience.

Education and Academy

The National Higher School of Computing graduates State Engineers in Informatics. The training is for five years spread out into three steps:

TWO YEARS OF PREPARATORY CLASSES CP:

The first year is dedicated to acquiring the foundational knowledge needed for developing an engineering mindset in computer science. This year is divided into two semesters:

first year ICP cycle preparatoire:

First Semester:

- **ALSDS (Algorithms and Static Data Structures):** This module covers the fundamental concepts of algorithms and data structures to develop skills in solving computational problems efficiently and optimally.
- **ARCH1 (Computer Architecture 1):** This course introduces the fundamental concepts of basic logic circuits, including Boolean algebra, as well as the study of combinational and sequential circuits, to understand the operation of digital systems and their role in computer architecture.
- **ANA1 (Analysis 1) and ALG1 (Algebra 1):** These modules provide the theoretical foundations necessary for understanding and applying fundamental mathematical concepts.
- **ELECT (Electricity):** This module covers the essential principles of electricity, including the laws governing currents and voltages.
- **SYS1 (Introduction to Operating Systems 1):** This course introduces the basic concepts of operating systems, with a particular focus on learning the Linux environment, its commands, file management, and essential features for managing and administering computer systems.
- **TEE (Techniques of Written Expression):** This module aims to develop writing and communication skills (in french).
- **BWEB (Office Tools and Web):** This module teaches essential office software tools and the basics of web development, allowing students to master technical skills useful for document management and simple website creation.

Second semester :

continues to build technical and scientific skills with more advanced modules:

- **ALSDD (Algorithms and Dynamic Data Structures):** This module deepens algorithmic concepts by introducing dynamic data structures, allowing the development of optimized solutions for more complex problems.
- **SYST2 (Introduction to Operating Systems 2):** In addition to continuing the study of operating systems, this course introduces assembly language, providing a better understanding of the interactions between hardware and software.
- **ANA2 (Analysis 2) and ALG2 (Algebra 2):** These modules extend the mathematical concepts from the first semester, exploring more complex theories necessary for modeling and solving scientific problems.
- **MECA (Mechanics of Point Mass):** This course covers the fundamental principles of classical mechanics, specifically the study of motion and forces applied to a material point.
- **ELECF1 (Fundamental Electronics 1):** This module introduces the fundamental concepts of electronics, including basic electronic components and their applications in circuits.
- **TEO (Oral Expression Techniques):** This course aims to develop oral communication skills in French, focusing on professional presentations and communication.
- **ANG1 (English 1):** This course focuses on computer-related vocabulary in English, aiming to improve students' language skills in the technical domain and enhance their ability to understand and use terms and expressions specific to computing in a professional context.

second year 2CP cycle preparatoire:

first semester :

marking the start of the second year, focuses on both technical and business-related subjects, broadening students' knowledge and preparing them for more advanced studies:

- **ECON (Business Economics):** This module introduces the principles of business economics, providing a foundation for understanding economic systems, business operations, and financial management in a professional context.
- **ANA3 (Mathematical Analysis 3):** This course builds on previous mathematical knowledge, focusing on advanced topics in mathematical analysis that are essential for solving complex problems in engineering and science.
- **ARCHI2 (Computer Architecture 2):** Following the first-year introduction, this module delves deeper into computer architecture, covering more advanced

concepts related to processors, memory hierarchies, and the internal functioning of computer systems.

- **ELECF2 (Fundamental Electronics 2):** This course continues the study of electronics, focusing on more complex circuits, components, and their applications in modern electronic systems.
- **ANG2 (English 2):** This module focuses on further developing technical English skills, emphasizing communication in professional and technical environments, particularly in the field of computer science.
- **ALG3 (Algebra 3):** This course advances students' understanding of algebraic structures and their applications in computing and problem-solving.
- **SFSD (File Structures and Data Structures):** This module focuses on the organization of data within file systems and the implementation of efficient data structures for managing large datasets.
- **PRST1 (Probability and Statistics 1):** This course introduces the basic principles of probability and statistics, essential for data analysis and decision-making processes in engineering.

second semester :

builds on the foundational knowledge from previous semesters while introducing new topics and practical experiences through teamwork:

- **ANG3 (English 3):** This module further develops students' technical English skills, focusing on communication in international professional and technical environments, particularly in the fields of computing and engineering.
- **LOGM (Mathematical Logic):** This course introduces the principles of mathematical logic, including formal systems, proof techniques, and their applications in computer science, laying the groundwork for theoretical aspects of computing.
- **OPTOE (Optics and Electromagnetic Waves):** This module covers the fundamental principles of optics and electromagnetic waves, essential for understanding how these phenomena are applied in technology, such as in communication systems and electronic devices.
- **PRST2 (Probability and Statistics 2):** Building on the concepts from the previous semester, this course delves deeper into statistical analysis and probabilistic models, with a focus on applications in data analysis and engineering decision-making.
- **POO (Object-Oriented Programming):** This course introduces object-oriented programming (OOP) concepts, teaching students how to design and implement efficient, reusable, and modular code using OOP principles.

- **ANA4 (Mathematical Analysis 4):** Continuing from previous analysis courses, this module covers more advanced topics in mathematical analysis, crucial for solving complex engineering and scientific problems.
- **SINF (Information Systems):** This module introduces the design and management of information systems, including data modeling, database management, and information system architectures, essential for managing and organizing data in business environments.
- **PRJP (2CPI Project):** In this team-based project, each group works on a specific theme in fields such as IoT, web development, or mobile development, providing hands-on experience in managing and completing a real-world project from concept to implementation.

About concours :

In the second year, each school (ESI, ESI SBA, ESTIN) organizes a competition, also known as the 2CP exam or concours, for access to the second cycle (ICS). Every year, the Ministry of Higher Education determines the number of available spots for the second cycle.

The top 80% of students in the ranking are directly admitted to the second cycle without having to take the exam. The remaining 20% of places are awarded to students who participate in the written exam, based on their ranking.

For example, at the end of the second year at ESI Algiers, students have two possibilities:

1. If their ranking is within the top 80%, they are directly admitted to the second cycle without having to sit the exam.
2. If their ranking is below the top 80%, they must take the written exam with students from ESI Algiers, ESI SBA, and ESTIN to compete for one of the remaining 20% of spots.

For the 2024/2025 season, out of 262 students in total, 209 students from ESI Algiers were directly admitted to the second cycle, with the cutoff grade being 11.33. Students who scored between 10.00 and 11.32 were eligible to take the concours as a second chance to pass into the next year's ICS. Students who scored below 10.00 failed the year. Out of those who took the concours, 37 students from ESI succeeded, while the others were transferred to other universities or schools.

For the 2023/2024 season, the cutoff grade for access to the concours was slightly lower, stopping at 11.23.

The second-cycle program lasts three years (six semesters), across all specialties:

- The first year (1CS) constitutes the common core.
- The second year (2CS) is dedicated to specialization.
- The third year (3CS) is dedicated to the final year project and Master (optional)

ONE YEAR OF COMMON CORE IN THE SECOND CYCLE (1CS), third 3rd year :

One year of common core in the second cycle (1CS) or Third year: Students succeeding in the competition will join the first year of common core. This year is divided into two semesters:

First semester modules :

- SYC1 – Centralized operating systems 1
- IGL – Introduction to Software Engineering
- English
- THP – Theory of programming languages
- ORGA – Analysis of organizations
- ANUM – digital analysis
- RO – Operational Research
- RES1 – Network 1

Second semester modules :

- CPRJ – Project Management
- SYC2 – Centralized operating systems 2
- SEC – Computer Security
- RES2 – Network 2
- ARCHI3 – Computer architecture 3
- Projet 1CS
- MCSI – information Systems Design Methodology
- English
- BDD – Database

TWO YEARS IN MAJOR (4 TH + 5 TH YEAR):

Two years in major (4TH + 5TH YEAR): The School offers to students four majors during the engineering cycle (SIT: Information and Technology Systems, SIQ: Informatics Systems, SIL: Software Engineering Systems and SID: Intelligent Systems and Data).

Students choose one of the majors by the end of the first year in the second cycle then they will be oriented based on their marks obtained during the academic year and the available seats.

SIT sit speciality: Information and Technology Systems :

It is a generalist engineering program in computer science with a focus on the field of information systems.

This training will allow students to equip themselves with broad knowledge and diverse skills related to the various professions in computer science, as well as solid expertise in the field of information systems and technologies (IS architectures, decision-support information systems, ERP, IS urbanization, quality assurance, etc.).

The knowledge they acquire in the various fields of computer science (software development, systems and networks, databases, information systems, etc.) at the end of their training will make them designers and developers capable of understanding the engineering of any computer system. The training they will receive in information systems and technology will give them the ability to analyze the needs of future users, align the company's information system with its strategic objectives, understand and suggest the most suitable organizational structures for a company or institution, manage an IT project, and more. It is divided into two semesters:

first semester modules of 2CS sit speciality :

- ANAD – Analyse De Données
- ASI – Analyse des Systèmes d'Information
- SIAD – Systèmes d'Aide à la Décision
- AQUA – Assurance Qualité
- MPSI – Conduite de changement
- BDA – SGBD et Base De Données Avancées
- TICO – TIC en Organisation

second semester modules of 2CS sit speciality:

- Projet 2CS
- FASI – File d'attente et Simulation
- ERP – Enterprise Ressource Planning
- SIC – Systèmes d'Information Coopératifs

- AL - Architecture Logicielle
- MSSSI-Management de la Sécurité des SI
- COFI - Comptabilité _ Finance
- Projet Sécurité Système d'Information

SIQ siq speciality: Informatics Systems :

It is a generalist engineering program in computer science with a focus on computer systems.

This training will allow students to acquire broad knowledge and diverse skills related to the various professions in computer science, as well as solid expertise in the field of computer systems (systems, networks, security, distributed computing, embedded systems, etc.).

The knowledge they will have acquired in the various fields of computer science (software development, systems and networks, databases, information systems, etc.) at the end of their training will make them designers and developers capable of understanding the engineering of any computer system. The training they will receive in computer systems will give them the ability to analyze complex problems, design, and develop IT solutions. These solutions require mastery of IT infrastructure (networks, systems, databases, cloud, etc.) and an engineering approach that allows for the design and use of efficient algorithms to solve the problems at hand. It is divided into two semesters:

first semester modules of 2CS siq speciality:

- VCL_2023
- COMPIL - Compilation
- RESA - Réseaux Avancés
- TPGO - Théorie de la Programmation et Graphes Orientés
- ANAD - Analyse De Données
- FAS - File d'Attente et Simulation
- STAGE

second semester modules of 2CS siq speciality:

- SSR - Sécurité Systèmes et Réseaux
- OPTM - Optimisation Combinatoire
- ALOG - Architectures logicielles
- BDA - SGBD et Base de Données Avancées

- Projet_2CS
- SYSR – Systèmes Répartis

SIL sil speciality: Software Engineering Systems :

It is a generalist engineering program in computer science with a focus on computer systems.

This training will equip students with broad knowledge and diverse skills related to the various professions in computer science, as well as solid expertise in software engineering (Agile methods, design patterns, HPC, mobile development, HCI, web development, etc.).

The knowledge they will have acquired in the various fields of computer science (software development, systems and networks, databases, information systems, etc.) at the end of their training will make them designers and developers capable of understanding the engineering of any computer system. The training they will receive in software engineering will give them the ability to develop software solutions that meet the needs of end users, master current technologies used in developing these solutions (Mobile, Web, Embedded, etc.), manage software development projects, and more. It is divided into two semesters:

first semester modules of 2CS sil speciality:

- WEB – Technologies De Développement Web
- VCL_2023
- TPGO
- OUTILS – Outils GL
- PDC – Patrons De Conception
- COMPIL – Compilation
- MAGIL – Méthodes Agiles
- ANAD – Analyse De Données
- Stage
- IHM – Interfaces Homme Machine

second semester modules of 2CS sil speciality:

- MBL2_Mobile 2
- BDM – Big Data mining
- BDA – SGBD et Bases de Données Avancées
- ENTP- Entrepreneuriat

- MBL1_Mobile1
- IHM - Interface Homme Machine
- MNG- management
- QLOG - Qualité Logicielle
- ALOG - Architectures Logicielles

SID sid speciality: Intelligent Systems and Data :

This is a general engineering training program in computer science, with a focus on data science and artificial intelligence.

This program will equip students with broad knowledge and diverse skills related to various fields of computer science, as well as strong expertise in data science and artificial intelligence.

The knowledge they will have acquired in different areas of computer science (software development, systems and networks, databases, information systems, etc.) by the end of their training will make them designers and developers capable of understanding the engineering of any computer system. The training they will receive in data science and artificial intelligence will give them the ability to analyze data and extract knowledge for decision-making, to design and implement systems based on artificial intelligence techniques. These require the design and use of algorithms for solving optimization and machine learning problems, with the goal of providing intelligent solutions to current issues in sectors as varied as healthcare, energy, smart cities, industry, transportation, agriculture, and education. . It is divided into two semesters:

first semester modules of 2CS sid speciality:

- CRP - Complexité et Résolution de Problème
- HPC - High Performance Computing
- TSG - Traitement du Signal
- ANAD - Analyse De Données
- BDA - SGBD et Base De Données Avancées
- MASD - Mathématiques avancées pour la science de données
- INFOVIS - Visualisation de l'information
- ML - Machine Learning
- SGOV - Smart Government

second semester modules of 2CS sid speciality:

- PMSS - Processus, Modèles et Simulation Stochastiques
- IRIAD - Informatique Répartie et Intelligence Artificielle distribuée

- NLP – Natural Language Processing
- TOAI – Techniques d’optimisation et Intelligence Artificielle
- BI – Business Intelligence
- RV – Réalité Virtuelle
- RCR – Représentation des Connaissances et Raisonnement
- IMN – Images Numériques

Informations about CSE (Scientifique Club of ESI) :

The Scientific Club of ESI (CSE) is one of the most active and prestigious student clubs at Algeria's leading IT school, the École Supérieure d'Informatique (ESI). Founded in 2008 the first club in ESI, its mission is to promote a culture of innovation, technical expertise, and collaboration among students, especially those studying computer science and related fields. Beyond its active role within the community, the CSE is an environment where competent profiles are formed thanks to the collaborative spirit that prevails, allowing students to practice, in a volunteer capacity, an activity similar to their future professional projects, with the possibility of learning from their mistakes. The CSE notably organizes an annual hackathon called Hack!T, workshops and training sessions on topics like software development, cybersecurity, and AI, while partnering with industry and other organizations to offer students internships, research projects, and opportunities to collaborate on innovative tech initiatives, thus creating a supportive community where students can collaborate, share knowledge, and build lasting professional connections.

CSE main events:

Hack!T: A hackathon that is the first and largest of its kind in Algeria, bringing together students to develop innovative technological solutions in a competitive and creative environment.

DataHack: The biggest datathon in Algeria, which focuses on data science and artificial intelligence, providing students the opportunity to tackle real-world data challenges.

CSE Around Algeria: A project where club members travel to different regions of the country to democratize access to computer science and technology.

Hack IN: An internal hackathon specifically designed for beginner students to gain hands-on experience and learn from their peers.

CSE's structure :

These are CSE's department along with their managers and assistants :

- **Président** : mohamed erraid djemai
- **Vice Président** : Anes hannachi
- **Secrétaire Générale (SG)**: [@Hiba](#)
- **Human Resources Manager (RH)**: [@AYOUB kihal](#)
- **Marketing Manager**: [@Guerrara Fatima Zohra](#)
- **PROD Lead**: [@Younes abdat](#)
- **Multimedia Manager + Assistant**: [@Hammar Soulaf](#) & [@AIJAMATINE Basma](#)
- **Design Manager + Assistant**: [@Fares](#) & [@ishraq](#)
- **Communication Co-Managers**: [@Benarmas Nadir](#) & [@BOULANOUAR MOHAMED](#)
- **Dev Manager + Assistant**: [@Ikram](#) & [@Dali braham mohamed](#)
- **AI Co-Managers**: [@Djouider Mokhtar Anes](#) & [@Badreddine Sayah](#)
- **Webmaster**: [@rami](#)
- **Relev Manager + Assistant**: [@Oussama kf](#) & [@nouha](#)
- **Relax Co-Managers**: [@Mohamed Adem](#) & [@Imed Mouffok](#)
- **Logistics Manager**: [@Benkradidja Rayane](#)
- **UI/UX Manager + Assistant**: [@Ghada](#) & [@Nesrine](#)
- **Formation Co-Managers**: [@Manel](#) & [@Nihad](#)
- **Project Co-Managers**: [@Mahmoud](#) & [@Anes](#)
- **Activities Co-Managers**: [@Benkerri Yacine](#) & [@Meriem Elhaouari](#)
- **Ex président** : Boussebata Issam

Questions about ESI and CSE :

What are the positive aspects of the school?

ESI Algiers offers a collaborative and supportive environment, with a strong focus on academic excellence. The clubs, like the CSE Club, provide great opportunities for students to develop technical skills, engage in events, and grow both personally and professionally.

What are the negative aspects of the school?

At times, the workload can be overwhelming, especially during exam periods or project deadlines, which can be challenging to manage alongside extracurricular activities.

How would you describe the environment at ESI?

The environment at ESI is motivational, enjoyable, and supportive. It encourages learning, collaboration, and personal growth, making it a positive space for students to thrive academically and socially.

What are the modules that I would struggle with the most?

It varies from person to person, but generally, all modules are interesting and manageable. With the right effort, you can do well in all of them!

Is getting good grades in the first semester possible or not?

Yes, it's definitely possible! Many students have achieved grades of 15+ in their first semester with consistent effort and focus.

What are some study methods and techniques that would enhance my grades?

Some effective study techniques include:

- **Active recall:** Testing yourself on the material instead of just reviewing it.
- **Spaced repetition:** Revisiting topics at increasing intervals to strengthen memory.
- **The Feynman technique:** Teaching concepts in simple terms to ensure you understand them deeply.
- **Pomodoro technique:** Breaking study sessions into focused intervals with short breaks to stay productive.
- **Practice problems:** Applying what you've learned through exercises and real-world examples.

These methods can help you retain information better and improve your performance.

What is the difference between lectures, tutorials, and practicals?

- **Lecture:** A session where you learn the core concepts of a subject, usually through a professor's explanation.
- **Tutorial:** A smaller session focused on practicing exercises and solving problems based on the lecture material.
- **Practical:** Hands-on sessions where you apply what you've learned, typically through programming or real-world tasks.

Will I need a computer in my first year?

Yes, having a computer is important for completing practical assignments (TPs) and programming tasks. It will also help with studying and coursework.

Do we need a high-performance laptop in the prépa cycle?

No, a standard laptop is usually sufficient. A machine with around 4-8 GB of RAM and an SSD should be enough to run most programs you'll need during the prépa cycle.

In which language are the classes and tests?

Most classes are conducted in English, while tests are offered in both French and English.

Is it true that as an ESI student you can't practice your hobbies/sport?

No, that's not true. You can definitely practice your hobbies and engage in sports; you just need to be organized with your time.

Is the elimination grade fixed or not?

No, the elimination grade is not fixed. It depends on the average grades of all students, calculated as 60% of the average of their grades.

Is it possible to join a club in the first year, and can you balance that with studies?

Yes, you can join clubs in your first year and still achieve good grades. You just need to prioritize your time effectively.

Does a graduate from ESI find a job right after finishing their studies or not?

Generally, most students from ESI are able to secure respectable jobs in various sectors right after graduation.

Is it possible to redo a year in the preparatory cycle?

Yes, it is possible to redo a year in the preparatory cycle.

How do exams and continuous assessments work in the preparatory cycle?

In the preparatory cycle, you have a week dedicated to exams for various modules, typically lasting between 1.5 to 2 hours. These exams cover all the material studied up to that point. Continuous assessments may also occur throughout the year to gauge your understanding.

What is the level of involvement of alumni in the life of the school?

Many alumni remain active in the computer science community, with some becoming researchers or professionals in the field. You can definitely reach out to them for help and support when needed.

How can students receive help for their projects or theses?

Students can seek help by approaching members of clubs or visiting the library, where they can find guidance and support from peers and resources.

Where can I find resources (exercise sheets, courses, exams, etc.)?

You can find resources on the website created by the CSE Club: tresor.cse.club.

What is the purpose of ESI Talents?

ESI Talents is the platform where students receive their grades and access important files and resources.

Is it possible to retake an exam if we have a justified absence?

Yes, you can retake an exam through a replacement exam (CR) if you have a justified absence.

What is the 1st-year discovery internship? Is it mandatory?

The 1st-year discovery internship is designed to give you insight into how an enterprise operates. It is not mandatory.

Can the clubs at ESI have a positive impact on my professional career?

Yes, clubs at ESI can have a significant positive impact on your professional career by helping you develop both soft and hard skills that are valuable in the workplace.

If we decide to do the discovery internship, is it up to us to choose the company, or does the school take care of it?

You will need to find the company for your internship on your own. The school does not assign companies.

Who developed CISSOU ?

CISSOU was developed by the CSE's development and artificial intelligence manager Sayah Badreddine inspired from the idea of the alumni: Yasmine Zaidi , Missoum Yousuf and Hamza Arab.

How can one join CSE or participate in CSE's events?

Any student from any field can join the CSE.

Just follow our announcements on our social media! We open registrations at the beginning of each year to recruit new members. As for our events, they are often open to all ESI students,

regardless of their level. Whether you're a beginner or an expert, you'll always find an activity or a project where you can get involved. We truly believe that everyone has something to contribute.

What distinguishes CSE from other scientific clubs in Algeria?

We have a spirit of constant innovation and renewal. For example, we impose an internal rule: never to organize more than three editions of the same external event. This forces us to reinvent our concepts and create new formats that surprise and engage the community. Additionally, we place a strong emphasis on the quality of the relationships we build with local and international partners, which gives our members access to unique opportunities.

What is ESI 101 ?

ESI 101 is an exclusive event for first-year students to kickstart their journey at ESI! We'll guide you through essential insights on student life, academic modules, and balancing studies with extracurriculars. From navigating courses to joining clubs, we've got everything you need for success from day one!

? Extra Questions & Answers about ESI

Q: What makes ESI different from other universities in Algeria?

A: Unlike standard universities, ESI is a **specialized Grande École** dedicated solely to computer science and engineering. It is extremely **selective**, admits only the top 0.1% of baccalaureate students, and offers a **state engineer diploma** after five years of intense training. Its curriculum combines **theory (maths, algorithms, systems)** with **hands-on projects** (IoT, AI, data science, software engineering). ESI also maintains strong links with industry, providing internships and career opportunities. This mix of selectivity, rigor, and real-world focus makes it stand out.

Q: How is student life at ESI outside of classes?

A: Life at ESI goes far beyond lectures and exams. The campus is buzzing with **student clubs, scientific events, hackathons, workshops, and competitions**. Many students join clubs like **CSE**, which help them build leadership and teamwork skills. Socially, ESI has a strong community vibe—students often form study groups, take part in sports, and organize cultural events. For many, these experiences are as valuable as academics because they provide **friendships, networks, and practical confidence**.

Q: What are some of the biggest challenges students face at ESI?

A: The **academic workload** is one of the most challenging aspects, especially during the preparatory years. Courses are highly demanding, and exams require deep understanding rather

than memorization. Some students also struggle with **time management**, balancing studies with extracurricular activities. For those coming from outside Algiers, there can be issues of **adaptation, homesickness, and stress**. However, many students overcome these difficulties by joining clubs, seeking peer support, and developing solid study habits.

Q: What kind of jobs can ESI graduates get after finishing their studies?

A: ESI graduates are highly sought after both in Algeria and internationally. Typical career paths include **software engineering, data science, artificial intelligence, cybersecurity, cloud computing, IT project management, systems engineering, and academia**. Many alumni work for top tech companies in Europe, Canada, or the Middle East, while others become **entrepreneurs and startup founders** in Algeria's growing tech ecosystem. Thanks to their solid technical foundation, ESI graduates adapt easily to different industries like finance, healthcare, energy, and telecommunications.

Q: How does the curriculum prepare students for the real world?

A: The program is **progressive**: students first build solid foundations in math, algorithms, and system basics. Then they gradually move into **applied areas like software engineering, AI, networks, and data science**. Every year, students work on **projects** that mimic real industry work, from small apps in the preparatory cycle to full-scale engineering projects in the last two years. The **final year project (PFE)** is often conducted with industry partners, giving students exposure to real challenges and professional environments.

Q: Can international students join ESI?

A: Historically, ESI (formerly CERI/INI) trained not only Algerians but also students from other African countries like **Tunisia, Mali, Cameroon, Senegal, and Mauritania**. Today, most places are reserved for Algerian students through the national baccalaureate system, but international students can sometimes be admitted through special programs, partnerships, or government scholarships.

? Extra Questions & Answers about CSE (Scientific Club of ESI)

Q: Why is CSE considered the most important student club at ESI?

A: CSE is more than just a club—it's the **heart of student innovation** at ESI. Founded in 2008, it was the **first student club at ESI**, and since then it has grown into the **largest and most active student scientific club in Algeria**. CSE is responsible for organizing **the country's first hackathon (Hack!T)**, as well as **DataHack, CSE Around Algeria, workshops, and national events**. It also created useful platforms like **Trésor (for resources)** and innovative apps like **CISSOU**. Its members often become

some of the **most skilled and influential students** on campus because they combine academic knowledge with practical, real-world experience.

Q: How does being a member of CSE benefit students personally and professionally?

A: Joining CSE provides a **double advantage**. Personally, it helps students **gain confidence, teamwork experience, leadership skills, and public speaking ability**. Professionally, members practice **coding, design, project management, and AI development** in environments similar to real-world jobs. They also meet industry professionals, work on collaborative projects, and often secure internships thanks to their involvement. Many alumni credit CSE as a **turning point in their careers** because it gave them both technical expertise and soft skills that employers value.

Q: What kind of big events does CSE organize?

A: CSE organizes some of the **most prestigious tech events in Algeria**:

- **Hack!T** – Algeria’s first hackathon, where students work 48h nonstop to develop innovative solutions.
- **DataHack** – the biggest datathon in Algeria, focusing on AI and real-world data challenges.
- **CSE Around Algeria** – members travel across the country to share tech knowledge and workshops with students in remote areas.
- **Hack IN** – a beginner-friendly hackathon to introduce new students to team-based innovation.
- **Workshops & Trainings** – covering topics like AI, cybersecurity, cloud, mobile, and design.

These events give students opportunities to test their knowledge, **network with professionals**, and **experience competition at a national level**.

Q: How is CSE structured internally?

A: CSE is managed like a **mini tech company**. It has a **president, vice president, and secretary general**, supported by departments for **HR, marketing, development, AI, multimedia, communication, design, logistics, training, projects, and external relations**. Each department has a **manager and assistants**, and members can join based on their interests. This structure allows CSE to run large-scale projects and events smoothly, while also giving students leadership experience.

Q: Is CSE only for advanced students?

A: No, **CSE welcomes everyone**, from complete beginners to experts. New students can join as early as their **first year**, even without prior coding or technical experience. The club provides **training sessions, mentoring, and beginner-friendly projects** so that everyone can learn and grow at their own pace. The philosophy of CSE is that **everyone has something to contribute**, whether it's coding, design, organization, or communication.

Q: What makes CSE different from other clubs in Algeria?

A: CSE has a **unique philosophy of innovation**: it avoids repeating the same external event more than three times, forcing members to constantly **reinvent and create new formats**. It also builds **strong partnerships** with local and international companies, giving students unique opportunities for internships and collaborations. Finally, its impact is **national**, not just limited to ESI—it spreads tech culture to other universities and regions through initiatives like **CSE Around Algeria**.

Q: Can participation in CSE help me get a job after graduation?

A: Absolutely. Recruiters often look for candidates who not only have **academic knowledge** but also **practical experience** and **soft skills**. By being active in CSE, students can showcase **real projects, teamwork, leadership, and innovation** on their CVs. Many ESI alumni say that CSE was the reason they stood out in interviews and landed positions in top companies or got accepted into international master's/PhD programs.

Q: Where can I find all official information about ESI? And where can I view my grades and personal information?

A: All official information about ESI can be found on the [official ESI website](#): ESI Official Website

To view your grades and track your academic progress, you can use the [Talent ESI](#) platform and the **Webetu Progress** app. These platforms allow you to access your grades, personal information, and academic records.

Q: Are there any great resources or old exams to help with revision? How can I revise and where can I find the old exams?

A: For ESI students, the primary platform to use is [Trésor ESI](#), created by the CSE. It offers a comprehensive collection of resources to support your studies, including old exams with solutions, tutorials (TDs), courses, and programs. This platform is an excellent tool to help you master the material and prepare for upcoming exams.

You can access all the necessary materials here: <https://tresor.cse.club/>

